

6. The beta decay of $^{14}_6\text{C}$ can be used for dating organic samples. The half-life of $^{14}_6\text{C}$ is 5730 years.
The percentage of $^{14}_6\text{C}$ atoms in a piece of wood from an old boat is found to have **decreased by** 30% since it was formed from living material.

(a) Estimate the age of the boat.

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(b) Calculate the activity of the $^{14}_6\text{C}$ after 7000 years as a percentage of its original activity.

[3]

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